

Answers and Discussion

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Part 1: Simple SQL Queries Begin!

This category deals with the basics of SQL. It covers select and where clauses, case expressions, unions, and a few other odds and ends. If you're already educated in SQL you will probably find these exercises fairly easy. If not, you should find them a good point to start learning for the more difficult categories ahead!

Link to practice: <https://pgexercises.com/questions/basic/>

Read the instructions and complete 12 questions below:

- [1] Retrieve everything from a table
- [2] Retrieve specific columns from a table
- [3] Control which rows are retrieved
- [4] Control which rows are retrieved - part 2
- [5] Basic string searches
- [6] Matching against multiple possible values
- [7] Classify results into buckets
- [8] Working with dates
- [9] Removing duplicates, and ordering results
- [10] Combining results from multiple queries
- [11] Simple aggregation
- [12] More aggregation

Example:

Question

How can you retrieve all the information from the cd facilities table?

Schema reminder

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
jondate	timestamp

facid	integer
memid	integer
starttime	timestamp
slots	integer

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
0	Tennis Court 1	5	25	10000	200
1	Tennis Court 2	5	25	8000	200
2	Badminton Court	0	15.5	4000	50
3	Table Tennis	0	5	320	10
4	Massage Room 1	35	80	4000	3000
5	Massage Room 2	35	80	4000	3000
6	Squash Court	3.5	17.5	5000	80
7	Snooker Table	0	5	450	15
8	Pool Table	0	5	400	15

Your Answer ✓

```
select *
from cd.facilities;
```

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
0	Tennis Court 1	5	25	10000	200
1	Tennis Court 2	5	25	8000	200
2	Badminton Court	0	15.5	4000	50
3	Table Tennis	0	5	320	10
4	Massage Room 1	35	80	4000	3000

Answers and Discussion

Part 2: Joins and Subqueries

This category deals primarily with a foundational concept in relational database systems: joining. Joining allows you to combine related information from multiple tables to answer a question. This isn't just beneficial for ease of querying: a lack of join capability encourages denormalisation of data, which increases the complexity of keeping your data internally consistent.

This topic covers inner, outer, and self joins, as well as spending a little time on subqueries (queries within queries)

Link to practice: <https://pgexercises.com/questions/joins/>

Read the instructions and complete the 8 questions below:

- [1] Retrieve the start times of members' bookings
- [2] Work out the start times of bookings for tennis courts
- [3] Produce a list of all members who have recommended another member
- [4] Produce a list of all members, along with their recommender
- [5] Produce a list of all members who have used a tennis court
- [6] Produce a list of costly bookings
- [7] Produce a list of all members, along with their recommender, using no joins.
- [8] Produce a list of costly bookings, using a subquery

Your Answer:

The results (SQL script and report) must be submitted to your GitHub account and share link here

Link: <https://github.com/pgthaotran/MIS-443---Q2-2025-2026-.git>

Below can as sample report:

Part 1: Simple SQL Queries Begin!

Question

How can you retrieve all the information from the cd.facilities table?

Schema reminder

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
0	Tennis Court 1	5	25	10000	200
1	Tennis Court 2	5	25	8000	200
2	Badminton Court	0	15.5	4000	50
3	Table Tennis	0	5	320	10
4	Massage Room 1	35	80	4000	3000
5	Massage Room 2	35	80	4000	3000
6	Squash Court	3.5	17.5	5000	80
7	Snooker Table	0	5	450	15
8	Pool Table	0	5	400	15

Your Answer

Hint

Help

Save

Run Query

```
select * from cd.facilities;
```

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
0	Tennis Court 1	5	25	10000	200
1	Tennis Court 2	5	25	8000	200
2	Badminton Court	0	15.5	4000	50
3	Table Tennis	0	5	320	10
4	Massage Room 1	35	80	4000	3000
5	Massage Room 2	35	80	4000	3000
6	Squash Court	3.5	17.5	5000	80
7	Snooker Table	0	5	450	15
8	Pool Table	0	5	400	15

Question

You want to print out a list of all of the facilities and their cost to members. How would you retrieve a list of only facility names and costs?

Schema reminder

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

name	membercost
Tennis Court 1	5
Tennis Court 2	5
Badminton Court	0
Table Tennis	0
Massage Room 1	35
Massage Room 2	35
Squash Court	3.5
Snooker Table	0
Pool Table	0

Your Answer

Hint

Help

Save

Run Query

```
select name, membercost
from cd.facilities
;
```

name	membercost
Tennis Court 1	5
Tennis Court 2	5
Badminton Court	0
Table Tennis	0

Question

How can you produce a list of facilities that charge a fee to members?

Schema reminder

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
0	Tennis Court 1	5	25	10000	200
1	Tennis Court 2	5	25	8000	200
4	Massage Room 1	35	80	4000	3000
5	Massage Room 2	35	80	4000	3000
6	Squash Court	3.5	17.5	5000	80

Your Answer

Hint

Help

Save

Run Query

```
select *
from cd.facilities
where membercost > 0
group by facid;
```

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
4	Massage Room 1	35	80	4000	3000
6	Squash Court	3.5	17.5	5000	80
0	Tennis Court 1	5	25	10000	200
1	Tennis Court 2	5	25	8000	200

Question

How can you produce a list of facilities that charge a fee to members, and that fee is less than 1/50th of the monthly maintenance cost? Return the facid, facility name, member cost, and monthly maintenance of the facilities in question.

Schema reminder

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

facid	name	membercost	monthlymaintenance
4	Massage Room 1	35	3000
5	Massage Room 2	35	3000

Your Answer

Hint

Help

Save

Run Query

```
select facid, name, membercost, monthlymaintenance
from cd.facilities
where membercost > 0 and membercost < monthlymaintenance/50
```

facid	name	membercost	monthlymaintenance
4	Massage Room 1	35	3000
5	Massage Room 2	35	3000

Question

How can you produce a list of all facilities with the word 'Tennis' in their name?

Schema reminder ▲

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
0	Tennis Court 1	5	25	10000	200
1	Tennis Court 2	5	25	8000	200
3	Table Tennis	0	5	320	10

Your Answer ✓

Hint Help Save Run Query

```
select *
from cd.facilities
where name like '%Tennis%'
;
```

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
0	Tennis Court 1	5	25	10000	200
1	Tennis Court 2	5	25	8000	200
3	Table Tennis	0	5	320	10

Question

How can you retrieve the details of facilities with ID 1 and 5? Try to do it without using the OR operator.

Schema reminder ▲

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
1	Tennis Court 2	5	25	8000	200
5	Massage Room 2	35	80	4000	3000

Your Answer ✓

Hint Help Save Run Query

```
select *
from cd.facilities
where facid in (1,5);
```

facid	name	membercost	guestcost	initialoutlay	monthlymaintenance
1	Tennis Court 2	5	25	8000	200
5	Massage Room 2	35	80	4000	3000

Question

How can you produce a list of facilities, with each labelled as 'cheap' or 'expensive' depending on if their monthly maintenance cost is more than \$100? Return the name and monthly maintenance of the facilities in question.

Schema reminder

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

name	cost
Tennis Court 1	expensive
Tennis Court 2	expensive
Badminton Court	cheap
Table Tennis	cheap
Massage Room 1	expensive
Massage Room 2	expensive
Squash Court	cheap
Snooker Table	cheap

Your Answer

HintHelpSaveRun Query

```
select name,
case
when monthlymaintenance > 100 then 'expensive'
else 'cheap'
end as cost
from cd.facilities;
```

name	cost
Tennis Court 1	expensive
Tennis Court 2	expensive
Badminton Court	cheap

Question

How can you produce a list of members who joined after the start of September 2012? Return the memid, surname, firstname, and joindate of the members in question.

Schema reminder

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

memid	surname	firstname	joindate
24	Sarwin	Ramnaresh	2012-09-01 08:44:42
26	Jones	Douglas	2012-09-02 18:43:05
27	Rumney	Henrietta	2012-09-05 08:42:35
28	Farrell	David	2012-09-15 08:22:05
29	Worthington-Smyth	Henry	2012-09-17 12:27:15
30	Purview	Millicent	2012-09-18 19:04:01
33	Tupperware	Hyacinth	2012-09-18 19:32:05
35	Hunt	John	2012-09-19 11:32:45

Your Answer

HintHelpSaveRun Query

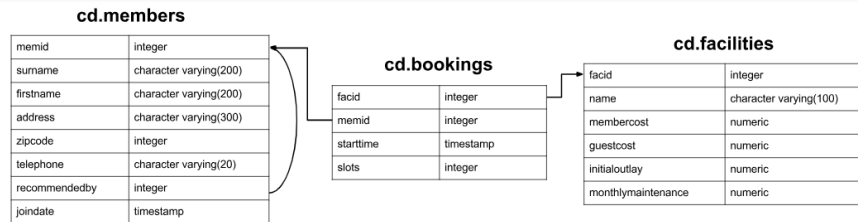
```
select memid, surname, firstname , joindate from cd.members
where joindate > '2012-09-01' ;
```

memid	surname	firstname	joindate
24	Sarwin	Ramnaresh	2012-09-01 08:44:42
26	Jones	Douglas	2012-09-02 18:43:05
27	Rumney	Henrietta	2012-09-05 08:42:35
28	Farrell	David	2012-09-15 08:22:05

Question

How can you produce an ordered list of the first 10 surnames in the members table? The list must not contain duplicates.

Schema reminder ▲



Expected Results

surname
Bader
Baker
Boothe
Butters
Coplin
Crumpet
Dare
Farrell
GUEST

Your Answer ✓

Hint Help Save Run Query

```
select DISTINCT surname
from cd.members order by surname limit 10;
```

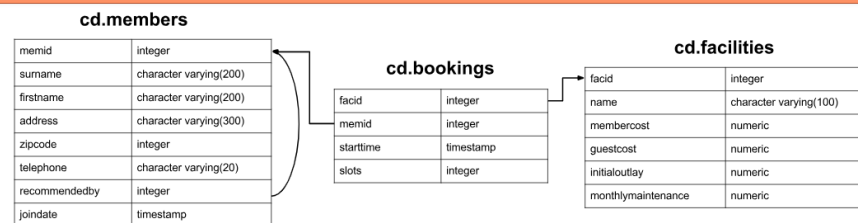
surname

Bader
Baker
Boothe
Butters

Question

You, for some reason, want a combined list of all surnames and all facility names. Yes, this is a contrived example :-). Produce that list!

Schema reminder ▲



Expected Results

surname
Tennis Court 2
Worthington-Smyth
Badminton Court
Pinker
Dare
Bader
Mackenzie
Crumpet

Your Answer ✓

Hint Help Save Run Query

```
select surname from cd.members
union
select name from cd.facilities;
```

surname

Hunt
Farrell
Tennis Court 2

Question

You'd like to get the signup date of your last member. How can you retrieve this information?

Schema reminder

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

latest

2012-09-26 18:08:45

Your Answer

HintHelpSaveRun Query

```
select MAX(joindate) as latest
from cd.members
```

latest

2012-09-26 18:08:45

Question

You'd like to get the first and last name of the last member(s) who signed up - not just the date. How can you do that?

Schema reminder

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

firstname	surname	joindate
Darren	Smith	2012-09-26 18:08:45

Your Answer

HintHelpSaveRun Query

```
select firstname, surname, joindate from cd.members
where joindate = (select max(joindate) from cd.members);
```

firstname	surname	joindate
Darren	Smith	2012-09-26 18:08:45

Part 2: Joins and Subqueries

Question

How can you produce a list of the start times for bookings by members named 'David Farrell'?

Schema reminder ▲

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

facid	integer
memid	integer
starttime	timestamp
slots	integer

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

starttime
2012-09-18 09:00:00
2012-09-18 17:30:00
2012-09-18 13:30:00
2012-09-18 20:00:00
2012-09-19 09:30:00
2012-09-19 15:00:00
2012-09-19 12:00:00
2012-09-20 15:30:00
2012-09-20 11:30:00

Your Answer ✓

Hint Help Save Run Query

```
select starttime from cd.bookings
left join cd.members on cd.members.memid = cd.bookings.memid
where surname = 'Farrell' and firstname = 'David';
```

starttime
2012-09-18 09:00:00
2012-09-18 13:30:00
2012-09-18 17:30:00
2012-09-18 20:00:00
2012-09-19 09:30:00
2012-09-19 12:00:00
2012-09-19 15:00:00
2012-09-20 11:30:00
2012-09-20 15:30:00

Question

How can you produce a list of the start times for bookings for tennis courts, for the date '2012-09-21'? Return a list of start time and facility name pairings, ordered by the time.

Schema reminder ▲

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

facid	integer
memid	integer
starttime	timestamp
slots	integer

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

start	name
2012-09-21 08:00:00	Tennis Court 1
2012-09-21 08:00:00	Tennis Court 2
2012-09-21 09:30:00	Tennis Court 1
2012-09-21 10:00:00	Tennis Court 2
2012-09-21 11:30:00	Tennis Court 2
2012-09-21 12:00:00	Tennis Court 1
2012-09-21 13:30:00	Tennis Court 1
2012-09-21 14:00:00	Tennis Court 2

Your Answer ✓

Hint Help Save Run Query

```
SELECT bks.starttime AS start, facts.name AS name
FROM cd.bookings AS bks
LEFT JOIN cd.facilities AS facts ON bks.facid = facts.facid
WHERE facts.name IN ('Tennis Court 1', 'Tennis Court 2')
AND bks.starttime >= '2012-09-21'
AND bks.starttime < '2012-09-22'
ORDER BY bks.starttime;
```

start	name
2012-09-21 08:00:00	Tennis Court 1
2012-09-21 08:00:00	Tennis Court 2
2012-09-21 09:30:00	Tennis Court 1
2012-09-21 10:00:00	Tennis Court 2

Question

How can you output a list of all members who have recommended another member? Ensure that there are no duplicates in the list, and that results are ordered by (surname, firstname).

Schema reminder ▲

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

firstname	surname
Florence	Bader
Timothy	Baker
Gerald	Butters
Jemima	Farrell
Matthew	Genting
David	Jones
Janice	Joplette
Millicent	Purview

Your Answer ✓

Hint Help Save Run Query

```
select distinct re.firstname as firstname, re.surname as surname
from cd.members mem
inner join cd.members re
on re.memid = mem.recommendedby
order by surname, firstname;
```

firstname	surname
Florence	Bader
Timothy	Baker
Gerald	Butters
Jemima	Farrell

Question

How can you output a list of all members, including the individual who recommended them (if any)? Ensure that results are ordered by (surname, firstname).

Schema reminder ▲

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

memfname	memsname	recfname	recsname
Florence	Bader	Ponder	Stibbons
Anne	Baker	Ponder	Stibbons
Timothy	Baker	Jemima	Farrell
Tim	Boothe	Tim	Rownam
Gerald	Butters	Darren	Smith
Joan	Coplin	Timothy	Baker
Erica	Crumpet	Tracy	Smith
Nancy	Dare	Janice	Joplette

Your Answer ✓

Hint Help Save Run Query

```
select mems.firstname as memfname, mems.surname as memsname, recs.firstname as recfname, recs.su
from cd.members mems
left outer join cd.members recs
on recs.memid = mems.recommendedby
order by memsname, memfname;
```

memfname	memsname	recfname	recsname
Florence	Bader	Ponder	Stibbons
Anne	Baker	Ponder	Stibbons
Timothy	Baker	Jemima	Farrell

Question

How can you produce a list of all members who have used a tennis court? Include in your output the name of the court, and the name of the member formatted as a single column. Ensure no duplicate data, and order by the member name followed by the facility name.

Schema reminder ▲

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

facid	integer
memid	integer
starttime	timestamp
slots	integer

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

member	facility
Anne Baker	Tennis Court 1
Anne Baker	Tennis Court 2
Burton Tracy	Tennis Court 1
Burton Tracy	Tennis Court 2
Charles Owen	Tennis Court 1
Charles Owen	Tennis Court 2
Darren Smith	Tennis Court 2

Your Answer ✓

Hint Help Save Run Query

```
select distinct cd.members.firstname || ' ' || cd.members.surname member, cd.facilities.name f
from cd.members
inner join cd.bookings
on cd.members.memid = cd.bookings.memid
inner join cd.facilities
on cd.bookings.facid = cd.facilities.facid
where cd.facilities.name in ('Tennis Court 1', 'Tennis Court 2')
order by member, facility;
```

member	facility
Anne Baker	Tennis Court 1
Anne Baker	Tennis Court 2
Burton Tracy	Tennis Court 1

Question

How can you produce a list of bookings on the day of 2012-09-14 which will cost the member (or guest) more than \$30? Remember that guests have different costs to members (the listed costs are per half-hour 'slot'), and the guest user is always ID 0. Include in your output the name of the member formatted as a single column, and the cost. Order by descending cost, and do not use any subqueries.

Schema reminder ▲

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

facid	integer
memid	integer
starttime	timestamp
slots	integer

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

member	facility	cost
GUEST GUEST	Massage Room 2	320
GUEST GUEST	Massage Room 1	160
GUEST GUEST	Massage Room 1	160
GUEST GUEST	Massage Room 1	160
GUEST GUEST	Tennis Court 2	150
Jemima Farrell	Massage Room 1	140
GUEST GUEST	Tennis Court 1	75

Your Answer ✓

Hint Help Save Run Query

```
SELECT
  mems.firstname || ' ' || mems.surname AS member,
  facs.name AS facility,
  CASE
    WHEN mems.memid = 0 THEN bks.slots * facs.guestcost
    ELSE bks.slots * facs.membercost
  END AS cost
FROM cd.members mems
JOIN cd.bookings bks ON mems.memid = bks.memid
```

member	facility	cost
GUEST GUEST	Massage Room 2	320
GUEST GUEST	Massage Room 1	160

Question

How can you output a list of all members, including the individual who recommended them (if any), without using any joins? Ensure that there are no duplicates in the list, and that each firstname + surname pairing is formatted as a column and ordered.

Schema reminder ▲

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

member	recommender
Anna Mackenzie	Darren Smith
Anne Baker	Ponder Stibbons
Burton Tracy	
Charles Owen	Darren Smith
Darren Smith	
David Farrell	
David Jones	Janice Joplette

Your Answer ✓

Hint

Help

Save

Run Query

```
SELECT DISTINCT
  mems.firstname || ' ' || mems.surname member,
  (SELECT rec.firstname || ' ' || rec.surname
   FROM cd.members rec
   WHERE rec.memid = mems.recommendedby) recommended_by
FROM cd.members mems
ORDER BY member;
```

member	recommended_by
Anna Mackenzie	Darren Smith
Anne Baker	Ponder Stibbons
Burton Tracy	

Question

The **Produce a list of costly bookings** exercise contained some messy logic: we had to calculate the booking cost in both the `WHERE` clause and the `CASE` statement. Try to simplify this calculation using subqueries. For reference, the question was:

How can you produce a list of bookings on the day of 2012-09-14 which will cost the member (or guest) more than \$30? Remember that guests have different costs to members (the listed costs are per half-hour 'slot'), and the guest user is always ID 0. Include in your output the name of the facility, the name of the member formatted as a single column, and the cost. Order by descending cost.

Schema reminder ▲

cd.members

memid	integer
surname	character varying(200)
firstname	character varying(200)
address	character varying(300)
zipcode	integer
telephone	character varying(20)
recommendedby	integer
joindate	timestamp

cd.bookings

facid	integer
memid	integer
starttime	timestamp
slots	integer

cd.facilities

facid	integer
name	character varying(100)
membercost	numeric
guestcost	numeric
initialoutlay	numeric
monthlymaintenance	numeric

Expected Results

member	facility	cost
GUEST GUEST	Massage Room 2	320
GUEST GUEST	Massage Room 1	160
GUEST GUEST	Massage Room 1	160
GUEST GUEST	Massage Room 1	160

Your Answer ✓

Hint

Help

Save

Run Query

```
SELECT member, facility, cost
FROM (
  SELECT
    CONCAT(mems.firstname, ' ', mems.surname) member,
    facs.name facility,
    CASE
      WHEN mems.memid = 0 THEN bks.slots * facs.guestcost
      ELSE bks.slots * facs.membercost
    END AS cost
  FROM cd.members mems
  JOIN cd.bookings bks ON bks.memid = mems.memid
  JOIN cd.facilities facs ON bks.facid = facs.facid
  WHERE bks.starttime <= '2012-09-14 00:00:00'
  AND bks.starttime >= '2012-09-14 00:00:00'
  AND bks.slots > 0
  ORDER BY cost DESC
)
```

member	facility	cost
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